Answer the following:

10 solution

$\text{Rate} = k[A]^2[B][L]^{3/2}$ -----(1)

(*i*) when $[L] = [4L]$

$\text{Rate} = k[A]^2[B][4L]^{3/2}$

$\text{Rate} = 8\left(k[A]^2[B][L]^{3/2}\right)$-----(2)

Comparing (1) and (2) ; rate is increased by 8 times.

(*ii*) when $[A] = [2A] \; and \; [B] = [2B]$

$\text{Rate} = k[2A]^2[2B][L]^{3/2}$

$\text{Rate} = 8\left(k[A]^2[B][L]^{3/2}\right)$-----(3)

Comparing (1) and (3) ; rate is increased by 8 times.

(*iii*) when $[A] = \left[\frac{A}{2}\right]$

$\text{Rate} = k\left[\frac{A}{2}\right]^2[B][L]^{3/2}$

$\text{Rate} = \left(\frac{1}{4}\right)\left(k[A]^2[B][L]^{3/2}\right)$-----(4)

Comparing (1) and (4) ; rate is reduced to ¼ times.

(*iv*) when $[A] = \left[\frac{A}{3}\right] \; and \; [L] = [4L]$

$\text{Rate} = k\left[\frac{A}{3}\right]^2[B][4L]^{3/2}$

$\text{Rate} = \left(\frac{8}{9}\right)\left(k[A]^2[B][L]^{3/2}\right)$-----(5)

Comparing (1) and (5) ; rate is reduced to 8/9 times.

11. solution

Let us consider the dimerisation of a monomer M

$2M \longrightarrow (M)_2$

$\text{Rate} = k[M]^n$

Given that n=2 and [M] = 0.05 mol L^{-1}

Rate = 7.5 X 10^{-3} mol $L^{-1}s^{-1}$

Answers.indd 251 3/29/2019 6:44:09 PM